



**STEEL AUTHORITY OF INDIA LIMITED
BHILAI STEEL PLANT
RESEARCH & CONTROL LAB (MECHANICAL TESTING LAB)**

Page No :B-1 of 1

IS : 2062



CM/L-0057534

MTL-II/QR-2

**TEST CERTIFICATE FOR HOT ROLLED MEDIUM AND HIGH TENSILE STRUCTURAL
STEEL**

Sold-to-Party:

BM BSO JAIPUR

S-16A LANDMARK 4TH FLOOR OPP:GUJARATI SAMAJ MAHAVIR

JAIPUR 302001

Ship-to-Party:

WHM JAIPUR,Jaipur.

T.C.No

: RCL/MTL/PLM/80931329

T.C. Date

: 24.03.2026

SALES ORDER NO

: 1100584217

DA No

: **80931329**

25

Wagon No / Trailer No

: ER26022610119

Process of Manufacture of Steel : Basic Oxygen Converter> Continuous Cast> Fully Killed Steel

Specification : IS 2062 E 350 BR : 2011 RA 2021

We certify that the material described below fully conforms to IS 2062 : 2011 RA 2021. Chemical composition and Mechanical properties of the product, as tested in accordance with the scheme of testing and inspection contained in the BIS certification mark Number CM/L 0057534 are as indicated against each Order No.

(Please refer to IS 2062 : 2011 RA 2021 for details of specification requirements)

TEST RESULTS

SECTION (NOMINAL SIZE)			Plate No / Pc. No.	Heat No	MECHANICAL PROPERTIES										Bend Test	M Dia	
THICK	WIDTH	LENGTH			YS	UTS	%E	%E2	CHARPY V NOTCH (L) Impact Values (J)								
mm	mm	mm			330 MPa Min	490 MPa Min	Min22(5.6 5Sqrt(S0))		I1	I2	I3	Iavg	°C				
25	2500	12000	3516561/1	413633	404	550	23	--	112	114	116	114	RT	OK	2.0 T		
				413633	397	548	24	--	--	--	--	--	--	OK	2.0 T		
25	2500	12000	3517806/1	413834	449	580	23	--	--	--	--	--	RT	OK	2.0 T		
				413834	457	587	22	--	90	102	98	97	RT	OK	2.0 T		
25	2500	12000	3517809/1	413839	406	530	24	--	80	86	88	85	RT	OK	2.0 T		
				413839	411	538	23	--	--	--	--	--	--	OK	2.0 T		
Heat No	CHEMICAL COMPOSITION (LADLE ANALYSIS)																
	C	S	P	Mn	Si	Al	Cu	Cr	Ni	Mo	Nb	V	Ti	N ₂	B	CE	
	%	%	%	%	%	%	%	%	%	%	%	%	%	ppm	%	%	
413633	0.16	0.013	0.020	1.45	0.35	0.021	< 0.02	< 0.015	< 0.015	< 0.015	< 0.010	0.052	< 0.005	68	< 0.0005	0.41	
413834	0.18	0.021	0.031	1.48	0.29	0.034	< 0.02	< 0.015	< 0.015	< 0.015	0.032	< 0.010	< 0.005	61	< 0.0005	0.43	
413839	0.15	0.025	0.026	1.37	0.25	0.023	< 0.02	< 0.015	< 0.015	< 0.015	0.037	< 0.010	< 0.005	66	< 0.0005	0.38	

Total Pieces Page / Sales Order : 3 / 3

Total Pieces / TC

: 13

The Material supplied conforms to the standard rolling and weight tolerances.,

CE calculated by formula: $[C + Mn/6 + (Cr + Mo + V)/5 + (Ni + Cu)/15]$.

RT- Room temperature as defined in the specification,

Size of impact test specimen: 10x10x55 mm(above 10mm thickness); 7.5x10x55 mm(upto 10mm thickness),

Note:** M = Mandrel Dia, T = Plate Thickness, Length is the rolling direction of the plate,

TC as per BSEN 10204 Type 3.1,

Validity unknown

Digitally signed by SAILJU
Date: 2026.03.24 18:55:59 IST
Reason: Authorized Signatory RCL,
BSP

Authorised Signatory

SAIL/ CMO/ BSO

Authorized Signatory

Research & Control Laboratory, Bhilai Steel Plant



STEEL AUTHORITY OF INDIA LIMITED
BHILAI STEEL PLANT
RESEARCH & CONTROL LAB (MECHANICAL TESTING LAB)

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S-16A LANDMARK 4TH FLOOR OPP:GUJARATI SAMAJ MAHAVIR

JAIPUR 302001

Ship-to-Party:

WHM JAIPUR,Jaipur.

T.C.No : RCL/MTL/PLM/80931329

T.C. Date : 24.03.2026

SALES ORDER NO : 1100584218

DA No : **80931329**

Wagon No / Trailer No : ER26022610119

Process of Manufacture of Steel : Basic Oxygen Converter> Continuous Cast> Fully Killed Steel

Specification : IS 2062 E 350 BR : 2011 RA 2021

We certify that the material described below fully conforms to IS 2062 : 2011 RA 2021. Chemical composition and Mechanical properties of the product, as tested in accordance with the scheme of testing and inspection contained in the BIS certification mark Number CM/L 0057534 are as indicated against each Order No.

(Please refer to IS 2062 : 2011 RA 2021 for details of specification requirements)

TEST RESULTS

SECTION (NOMINAL SIZE)			Plate No / Pc. No.	Heat No	MECHANICAL PROPERTIES										Bend Test	M Dia
THICK	WIDTH	LENGTH			YS	UTS	%E	%E2	CHARPY V NOTCH (L) Impact Values (J)							
mm	mm	mm			350 MPa Min	490 MPa	Min22(5.6 5Sqrt(S0))		I1	I2	I3	Iavg	°C			
16	2500	12000	3518745/2	413676	430	552	25	--	98	102	94	98	RT	OK	2.0 T	
				413676	441	556	24	--	--	--	--	--	--	OK	2.0 T	
16	2500	12000	3518753/1	413676	--	--	--	--	--	--	--	--	--	--	--	
16	2500	12000	3518753/2	413676	--	--	--	--	--	--	--	--	--	--	--	
16	2500	12000	3518754/1	413676	--	--	--	--	--	--	--	--	--	--	--	
Heat No	CHEMICAL COMPOSITION (LADLE ANALYSIS)															
	C	S	P	Mn	Si	Al	Cu	Cr	Ni	Mo	Nb	V	Ti	N ₂	B	CE
	%	%	%	%	%	%	%	%	%	%	%	%	%	ppm	%	%
413676	0.17	0.022	0.025	1.49	0.31	0.016	< 0.02	< 0.015	< 0.015	< 0.015	< 0.010	0.044	< 0.005	69	< 0.0005	0.43

Total Pieces Page / Sales Order : 4 / 4

Total Pieces / TC

: 13

The Material supplied conforms to the standard rolling and weight tolerances.,

CE calculated by formula: $[C + Mn/6 + (Cr + Mo + V)/5 + (Ni + Cu)/15]$,

RT- Room temperature as defined in the specification,

Size of impact test specimen:10x10x55 mm(above 10mm thickness) ; 7.5x10x55 mm(upto 10mm thickness),

Note:**M = Mandrel Dia, T = Plate Thickness, Length is the rolling direction of the plate,

TC as per BSEN 10204 Type 3.1,

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Research & Control Laboratory, Bhilai Steel Plant



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RESEARCH & CONTROL LAB (MECHANICAL TESTING LAB)

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Sold-to-Party:

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JAIPUR 302001

Ship-to-Party:

WHM JAIPUR,Jaipur.

T.C.No : RCL/MTL/PLM/80931329

T.C. Date : 24.03.2026

SALES ORDER NO : 1100584216

DA No : 80931329

Wagon No / Trailer No : ER26022610119

Process of Manufacture of Steel : Basic Oxygen Converter> Continuous Cast> Fully Killed Steel

Specification : IS 2062 E 350 BR : 2011 RA 2021

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TEST RESULTS

SECTION (NOMINAL SIZE)			Plate No / Pc. No.	Heat No	MECHANICAL PROPERTIES										Bend Test	M Dia	
THICK	WIDTH	LENGTH			YS	UTS	%E	%E2	CHARPY V NOTCH (L) Impact Values (J)								
mm	mm	mm			330 MPa Min	490 MPa	Min22(5.6 5SqrtS0)		I1	I2	I3	lavg	°C				
20	2500	12000	3517595/1	413215	410	545	23	--	88	90	102	93	RT	OK	2.0 T		
				413215	422	546	24	--	--	--	--	--	--	OK	2.0 T		
20	2500	12000	3517597/2	413215	--	--	--	--	--	--	--	--	--	--	--		
20	2500	12000	3517580/2	413225	430	555	24	--	90	98	86	91	RT	OK	2.0 T		
				413225	425	546	24	--	--	--	--	--	--	OK	2.0 T		
20	2500	12000	3517582/2	413225	--	--	--	--	--	--	--	--	--	--	--		
20	2500	12000	3518652/1	413331	447	566	25	--	116	110	118	115	RT	OK	2.0 T		
				413331	432	568	23	--	--	--	--	--	--	OK	2.0 T		
20	2500	12000	3518656/2	413331	--	--	--	--	--	--	--	--	--	--	--		
Heat No	CHEMICAL COMPOSITION (LADLE ANALYSIS)																
	C	S	P	Mn	Si	Al	Cu	Cr	Ni	Mo	Nb	V	Ti	N ₂	B	CE	
	%	%	%	%	%	%	%	%	%	%	%	%	%	ppm	%	%	
413215	0.17	0.020	0.023	1.40	0.28	0.030	< 0.02	< 0.015	< 0.015	< 0.015	< 0.010	0.046	< 0.005	65	< 0.0005	0.41	
413225	0.17	0.018	0.020	1.37	0.28	0.030	< 0.02	< 0.015	< 0.015	< 0.015	< 0.010	0.051	< 0.005	64	< 0.0005	0.41	
413331	0.17	0.024	0.018	1.40	0.28	0.033	< 0.02	< 0.015	< 0.015	< 0.015	< 0.010	0.051	< 0.005	66	< 0.0005	0.41	

Total Pieces Page / Sales Order : 6 / 6

Total Pieces / TC

: 13

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